

Quiz Answer Keys

Quiz 7 Answers

Q1. In data analysis, what process involves converting continuous numerical data into discrete intervals or bins?

■ Answer: (a) Binning

Q2. In machine learning, which ensemble method combines multiple decision trees by training them on different subsets of the training data and averaging their predictions to improve model accuracy and reduce overfitting?

■ Answer: (a) Random Forest

Q3. How can you save a Numpy array to a file?

■ Answer: (a) `numpy.save()`

Q4. In deep learning, what type of neural network architecture consists of multiple layers of interconnected neurons, including input, hidden, and output layers, where information flows only in one direction from input to output?

■ Answer: (c) MLP Feed Forward Neural Network

Q5. What does Bayesian probability involve?

■ Answer: (c) Calculating probabilities based on prior knowledge and observations

Q6. How can you create a Numpy array from a Python list?

■ Answer: (b) `numpy.array()`

Q7. Which ensemble method combines weak learners sequentially, where each subsequent learner focuses more on misclassified data from the previous ones, aiming to create a strong learner?

■ Answer: (d) Boosting

Q8. In Microsoft Azure, what networking solution allows you to isolate and secure your cloud resources by creating private network connections and controlling traffic flow between virtual machines and services?

■ Answer: (c) Azure Virtual Networks

Q9. In machine learning, what term refers to a situation where one class has significantly more instances than other classes in a dataset?

■ Answer: (d) Dataset imbalance

Q10. When developing an application that requires translating text from one language to another, which Azure service provides language translation capabilities and supports a wide range of languages for translation?

■ Answer: (d) Azure Translator

Q11. Which type of loop is suitable when the number of iterations is known beforehand?

■ Answer: (c) For loop

Q12. What is inheritance in object-oriented programming?

■ Answer: (a) It allows a class to inherit properties and methods from another class

Q13. In the Bag-of-Words (BoW) model, how is a text represented as a feature vector for NLP tasks?

■ Answer: (b) By counting the frequency of words in the text

Q14. What are the fundamental building blocks of programming that manipulate data?

■ Answer: (a) Operators

Q15. When creating an Azure AI resource, which service in the Azure portal allows you to provision and manage AI capabilities, such as cognitive services and machine learning models, for your applications?

■ **Answer: (d) Azure Cognitive Services**

Q16. When building a solution that involves converting spoken language into written text and performing speech-to-text transcription, which Azure service offers automatic speech recognition capabilities?

■ **Answer: (c) Azure Speech Service**

Quiz 8 Answers

Q1. When developing an application that requires custom image classification and object detection, which Azure Cognitive Service provides a user-friendly platform to train and deploy custom models for various visual recognition tasks?

■ **Answer: (b) Custom Vision**

Q2. What does polymorphism in OOP refer to?

■ **Answer: (d) Treating different class objects as instances of a common superclass**

Q3. If you're building a speech solution that involves converting spoken language into text, performing text-to-speech synthesis, and enabling speaker recognition, which Azure Cognitive Services offering is best suited?

■ **Answer: (c) Azure Speech Service**

Q4. In NLP, what is the process of converting text to a consistent format, such as converting all characters to lowercase and removing special characters?

■ **Answer: (d) Text normalization**

Q5. When creating an Azure AI resource, which service in the Azure portal allows you to provision and manage AI capabilities?

■ **Answer: (d) Azure Cognitive Services**

Q6. Which data structure is used to store an unordered collection of unique elements in Python?

■ **Answer: (b) Sets**

Q7. In natural language processing, what is the process of assigning grammatical tags to words in a text based on their roles in a sentence?

■ **Answer: (a) POS tagging**

Q8. In natural language processing, what is the process of reducing a word to its base or root form called, by removing suffixes or prefixes?

■ **Answer: (b) Stemming**

Q9. In supervised machine learning, which component represents the function that the model is trying to learn and approximate?

■ **Answer: (b) Hypothesis**

Q10. What are the fundamental building blocks of programming that manipulate data?

■ **Answer: (a) Operators**

Q11. What is the purpose of using keyword arguments in function calls?

■ **Answer: (b) To assign a name to an argument**

Q12. What is the purpose of software installation tools like Anaconda and VSCode in AI development?

■ **Answer: (b) To install and manage necessary programming environments**

Q13. What is the primary application of Convolutional Neural Networks (CNNs)?

■ **Answer: (c) Image and video analysis**

Q14. Which type of words are commonly removed from text during pre-processing due to their high frequency and lack of significant meaning?

■ **Answer: (b) Stopwords**

Q15. If you're developing an AI solution that involves extracting meaningful insights from unstructured text data, which Azure service is best suited?

■ **Answer: (a) Azure Cognitive Services**

Q16. What does the term 'probability' refer to in statistics?

■ **Answer: (a) The ratio of successful outcomes to the total possible outcomes**

Q17. When building a solution that provides personalized recommendations, which Azure service is best suited?

■ **Answer: (b) Azure Personalizer**

Q18. In logistic regression, what type of output does the model produce?

■ **Answer: (a) Probability scores**

Q19. When building a decision support solution that involves analyzing large datasets, which tool is best suited?

■ **Answer: (d) Microsoft Power BI**

Q20. What is the purpose of software installation tools like Anaconda and VSCode?

■ **Answer: (b) To install and manage necessary programming environments**

Quiz 9 Answers

Q1. In natural language processing, what is the process of assigning grammatical tags to words in a text based on their roles in a sentence?

■ **Answer: (a) POS tagging**

Q2. What is type casting in Numpy used for?

■ **Answer: (b) Changing the data type of an array**

Q3. When developing an application that requires understanding and processing natural language input, which Azure service is best suited?

■ **Answer: (a) Azure Language Understanding**

Q4. In machine learning, which algorithm creates a tree-like model of decisions and their possible consequences?

■ **Answer: (b) Decision Tree**

Q5. Which term in NLP refers to a curve that plots the performance of a model?

■ **Answer: (c) Learning curve**

Q6. In neural networks, what are the two main steps involved in the training process?

■ **Answer: (b) Forward and Backward Passes**

Q7. When planning for an AI solution that aligns with Responsible AI principles, what should be prioritized?

■ **Answer: (b) Explainability and Transparency**

Q8. In the classical machine learning pipeline, what step involves transforming raw data?

■ **Answer: (b) Data preprocessing**

Q9. In machine learning, which task involves predicting discrete class labels as output?

■ **Answer: (b) Classification**

Q10. In object-oriented programming, what are instance variables?

■ **Answer: (b) Variables that store data unique to each instance of a class**

Q11. What does multiple inheritance in OOP involve?

■ **Answer: (d) Inheriting from multiple parent classes**

Q12. When configuring security for Microsoft Azure AI services, which service is used?

■ **Answer: (d) Azure Active Directory**

Q13. Which evaluation metric is commonly used for binary classification problems?

■ **Answer: (b) Precision**

Q14. How can you convert a user's input into a numerical data type?

■ **Answer: (b) Through the input() function and type casting**

Q15. In cloud services, what are used as credentials to securely authenticate?

■ **Answer: (a) API keys**

Q16. In logistic regression for multiclass classification, what type of decision boundary is used?

■ **Answer: (b) Linear decision boundary**